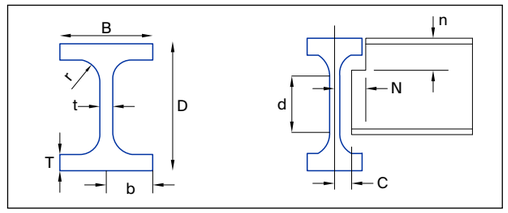
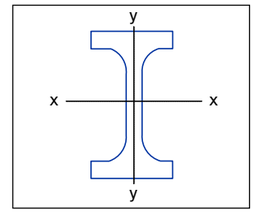
# 2.2
# Wide Flange Beams (HE)




| Dimensions | EN10365 |
|---|---|
| Specification | EN10025 |
| Size Range | HE 100 AA to HE 1000 x 584 |




| Section designation | Dimensions | | | | | | | | | | | | | Properties | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Mass per Metre | Depth of Section | Width of Section | Thickness | | Root | Depth between Fillets | Area of Section | Ratios for Local Buckling | | Dimensions for Detailing | | | Surface Area | Second Moment of Area | | Radius of Gyration | | Elastic Modulus | | Plastic Modulus | | Buckling Parameter | Torsional Index | Warping Constant | Torsional Constant |
| | | | | Flange | Web | | | | Flange | Web | End clearance | Notch | | Per metre | Axis x-x | Axis y-y | Axis x-x | Axis y-y | Axis x-x | Axis y-y | Axis x-x | Axis y-y | | | | |
| | kg/m | D mm | B mm | T mm | t mm | r mm | d mm | A cm² | b/T | d/t | C mm | N mm | n mm | m² | cm⁴ | cm⁴ | cm | cm | cm³ | cm³ | cm³ | cm³ | u | x | H dm⁶ | J cm⁴ |
| **HE 500 B** | 187 | 500.0 | 300.0 | 28.0 | 14.5 | 27.0 | 390.0 | 239 | 26.9 | 4.13 | 9 | 154 | 56 | 2.12 | 107,000 | 12,600 | 21.2 | 7.27 | 4,290 | 842 | 4,820 | 1,290 | 8.065 | 17.6 | 7.02 | 548 |
| **HE 500 M** | 270 | 524.0 | 306.0 | 40.0 | 21.0 | 27.0 | 390.0 | 344 | 18.6 | 2.89 | 13 | 154 | 68 | 2.18 | 162,000 | 19,200 | 21.7 | 7.46 | 6,180 | 1,250 | 7,090 | 1,930 | 8.157 | 12.9 | 11.2 | 1,540 |
| **HE 550 AA** | 120 | 522.0 | 300.0 | 15.0 | 11.5 | 27.0 | 438.0 | 153 | 38.1 | 7.82 | 8 | 156 | 42 | 2.17 | 72,900 | 6,770 | 21.8 | 6.65 | 2,790 | 451 | 3,130 | 699 | 7.708 | 31.5 | 4.34 | 127 |
| **HE 550 A** | 166 | 540.0 | 300.0 | 24.0 | 12.5 | 27.0 | 438.0 | 212 | 35.0 | 4.86 | 8 | 154 | 52 | 2.21 | 112,000 | 10,800 | 23.0 | 7.15 | 4,150 | 721 | 4,620 | 1,110 | 8.031 | 22.4 | 7.19 | 360 |
| **HE 550 B** | 199 | 550.0 | 300.0 | 29.0 | 15.0 | 27.0 | 438.0 | 254 | 29.2 | 3.98 | 10 | 154 | 56 | 2.22 | 137,000 | 13,100 | 23.2 | 7.17 | 4,970 | 872 | 5,590 | 1,340 | 8.045 | 19.0 | 8.86 | 610 |
| **HE 550 M** | 278 | 572.0 | 306.0 | 40.0 | 21.0 | 27.0 | 438.0 | 354 | 20.9 | 2.89 | 13 | 154 | 68 | 2.28 | 198,000 | 19,200 | 23.6 | 7.35 | 6,920 | 1,250 | 7,930 | 1,940 | 8.135 | 14.3 | 13.5 | 1,560 |
| **HE 600 AA** | 129 | 571.0 | 300.0 | 15.5 | 12.0 | 27.0 | 486.0 | 164 | 40.5 | 7.55 | 8 | 154 | 44 | 2.27 | 91,900 | 6,990 | 23.7 | 6.53 | 3,220 | 466 | 3,620 | 724 | 7.651 | 33.8 | 5.38 | 142 |
| **HE 600 A** | 178 | 590.0 | 300.0 | 25.0 | 13.0 | 27.0 | 486.0 | 226 | 37.4 | 4.66 | 9 | 154 | 52 | 2.31 | 141,000 | 11,300 | 25.0 | 7.05 | 4,790 | 751 | 5,350 | 1,160 | 8.001 | 23.8 | 8.98 | 407 |
| **HE 600 B** | 212 | 600.0 | 300.0 | 30.0 | 15.5 | 27.0 | 486.0 | 270 | 31.4 | 3.84 | 10 | 154 | 58 | 2.32 | 171,000 | 13,500 | 25.2 | 7.08 | 5,700 | 902 | 6,420 | 1,390 | 8.015 | 20.3 | 11.0 | 677 |
| **HE 600 M** | 285 | 620.0 | 305.0 | 40.0 | 21.0 | 27.0 | 486.0 | 364 | 23.1 | 2.88 | 13 | 152 | 68 | 2.37 | 237,000 | 19,000 | 25.6 | 7.22 | 7,660 | 1,240 | 8,770 | 1,930 | 8.088 | 15.8 | 15.9 | 1,570 |
| **HE 600 x 337** | 337 | 632.0 | 310.0 | 46.0 | 25.5 | 27.0 | 486.0 | 429 | 19.1 | 2.51 | 15 | 154 | 74 | 2.41 | 283,000 | 22,900 | 25.7 | 7.31 | 8,960 | 1,480 | 10,400 | 2,310 | 8.123 | 13.9 | 19.6 | 2,440 |
| **HE 600 x 399** | 399 | 648.0 | 315.0 | 54.0 | 30.0 | 27.0 | 486.0 | 508 | 16.2 | 2.14 | 17 | 154 | 82 | 2.45 | 345,000 | 28,300 | 26.0 | 7.46 | 10,600 | 1,800 | 12,500 | 2,810 | 8.201 | 12.0 | 24.8 | 3,930 |
| **HE 650 AA** | 138 | 620.0 | 300.0 | 16.0 | 12.5 | 27.0 | 534.0 | 176 | 42.7 | 7.30 | 8 | 154 | 44 | 2.37 | 114,000 | 7,220 | 25.5 | 6.41 | 3,680 | 481 | 4,160 | 751 | 7.594 | 35.9 | 6.57 | 158 |
| **HE 650 A** | 190 | 640.0 | 300.0 | 26.0 | 13.5 | 27.0 | 534.0 | 242 | 39.6 | 4.47 | 9 | 154 | 54 | 2.41 | 175,000 | 11,700 | 26.9 | 6.97 | 5,470 | 782 | 6,140 | 1,200 | 7.966 | 25.2 | 11.0 | 458 |
| **HE 650 B** | 225 | 650.0 | 300.0 | 31.0 | 16.0 | 27.0 | 534.0 | 286 | 33.4 | 3.71 | 10 | 152 | 58 | 2.42 | 211,000 | 14,000 | 27.1 | 6.99 | 6,480 | 932 | 7,320 | 1,440 | 7.978 | 21.6 | 13.4 | 749 |
| **HE 650 M** | 293 | 668.0 | 305.0 | 40.0 | 21.0 | 27.0 | 534.0 | 374 | 25.4 | 2.88 | 13 | 152 | 68 | 2.47 | 282,000 | 19,000 | 27.5 | 7.13 | 8,430 | 1,240 | 9,660 | 1,940 | 8.048 | 17.2 | 18.6 | 1,580 |
| **HE 650 x 343** | 343 | 680.0 | 309.0 | 46.0 | 25.0 | 27.0 | 534.0 | 438 | 21.4 | 2.50 | 15 | 152 | 74 | 2.50 | 334,000 | 22,700 | 27.6 | 7.21 | 9,820 | 1,470 | 11,400 | 2,300 | 8.084 | 15.2 | 22.7 | 2,440 |
| **HE 650 x 407** | 407 | 696.0 | 314.0 | 54.0 | 29.5 | 27.0 | 534.0 | 519 | 18.1 | 2.13 | 17 | 154 | 82 | 2.54 | 405,000 | 28,000 | 28.0 | 7.35 | 11,600 | 1,780 | 13,600 | 2,800 | 8.160 | 13.2 | 28.7 | 3,930 |
| **HE 700 AA** | 150 | 670.0 | 300.0 | 17.0 | 13.0 | 27.0 | 582.0 | 191 | 44.8 | 6.85 | 9 | 154 | 44 | 2.47 | 143,000 | 7,670 | 27.3 | 6.34 | 4,260 | 512 | 4,840 | 800 | 7.562 | 37.3 | 8.16 | 186 |
| **HE 700 A** | 204 | 690.0 | 300.0 | 27.0 | 14.5 | 27.0 | 582.0 | 260 | 40.1 | 4.29 | 9 | 154 | 54 | 2.50 | 215,000 | 12,200 | 28.8 | 6.84 | 6,240 | 812 | 7,030 | 1,260 | 7.897 | 26.5 | 13.4 | 522 |
| **HE 700 B** | 241 | 700.0 | 300.0 | 32.0 | 17.0 | 27.0 | 582.0 | 306 | 34.2 | 3.58 | 11 | 152 | 60 | 2.52 | 257,000 | 14,400 | 29.0 | 6.87 | 7,340 | 963 | 8,330 | 1,500 | 7.912 | 22.8 | 16.1 | 839 |
| **HE 700 M** | 301 | 716.0 | 304.0 | 40.0 | 21.0 | 27.0 | 582.0 | 383 | 27.7 | 2.86 | 13 | 152 | 68 | 2.56 | 329,000 | 18,800 | 29.3 | 7.01 | 9,200 | 1,240 | 10,500 | 1,930 | 7.991 | 18.7 | 21.4 | 1,600 |
| **HE 700 x 352** | 352 | 728.0 | 308.0 | 46.0 | 25.0 | 27.0 | 582.0 | 449 | 23.3 | 2.49 | 15 | 152 | 74 | 2.59 | 390,000 | 22,500 | 29.5 | 7.08 | 10,700 | 1,460 | 12,400 | 2,290 | 8.027 | 16.5 | 26.0 | 2,450 |
| **HE 700 x 418** | 418 | 744.0 | 313.0 | 54.0 | 29.5 | 27.0 | 582.0 | 532 | 19.7 | 2.13 | 17 | 152 | 82 | 2.63 | 472,000 | 27,800 | 29.8 | 7.22 | 12,700 | 1,770 | 14,800 | 2,800 | 8.100 | 14.3 | 32.8 | 3,960 |
| **HE 800 AA** | 172 | 770.0 | 300.0 | 18.0 | 14.0 | 30.0 | 674.0 | 218 | 48.1 | 6.28 | 9 | 154 | 48 | 2.66 | 209,000 | 8,130 | 30.9 | 6.10 | 5,430 | 542 | 6,220 | 857 | 7.424 | 40.2 | 11.4 | 243 |